

#23070521141, Shivansh Lohani

!pip install tensorflow

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```

```
from tensorflow import keras
from tensorflow.keras import layers
```

```
model = keras.Sequential([
    layers.Dense(units=32, activation='relu', input_shape=(10,)),
    layers.Dense(units=16, activation='relu'),
    layers.Dense(units=1, activation='sigmoid')
])
```

```
model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 32)	352
dense_1 (Dense)	(None, 16)	528
dense_2 (Dense)	(None, 1)	17

```
Total params: 897 (3.50 KB)
Trainable params: 897 (3.50 KB)
Non-trainable params: 0 (0.00 B)
```

```
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

```
print("Model compiled successfully!")
```

```
import numpy as np
x_train = np.random.rand(100, 10).astype(np.float32)
y_train = np.random.randint(0, 2, size=(100, 1)).astype(np.float32)
```

```
print("\nStarting model training...")
history = model.fit(x_train, y_train, epochs=5, batch_size=32, verbose=1)
```

```
print("\nModel training complete.")
```

Model compiled successfully!

Starting model training...

Epoch 1/5

4/4  2s 14ms/step - accuracy: 0.5700 - loss: 0.6880

Epoch 2/5

4/4  0s 18ms/step - accuracy: 0.6300 - loss: 0.6849

Epoch 3/5

4/4  0s 13ms/step - accuracy: 0.6300 - loss: 0.6831

Epoch 4/5

4/4  0s 14ms/step - accuracy: 0.6300 - loss: 0.6813

Epoch 5/5

4/4  0s 15ms/step - accuracy: 0.6200 - loss: 0.6801

Model training complete.